

Command Line Basics

John Garza

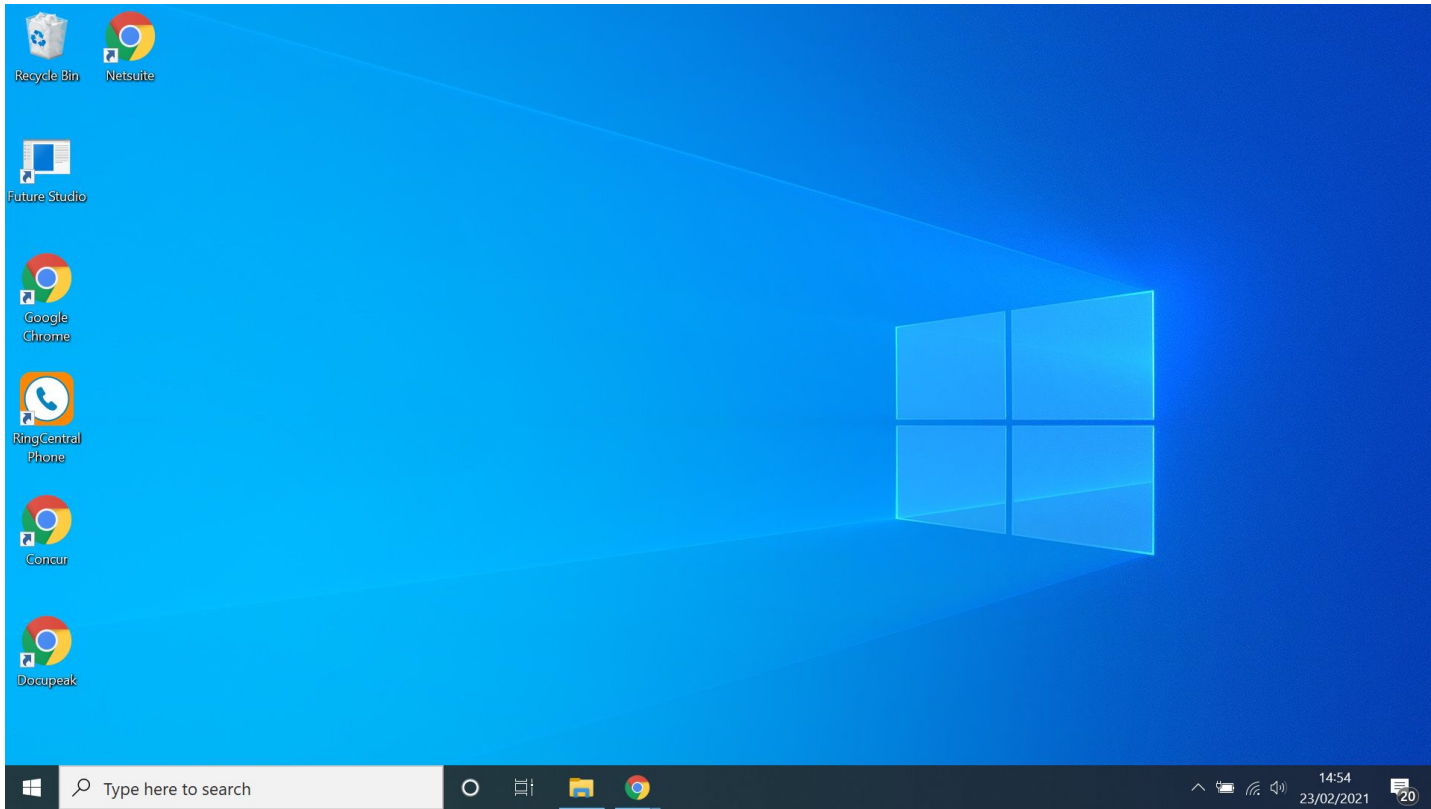
Follow Along

<https://sandbox.bio/tutorials/terminal-basics>

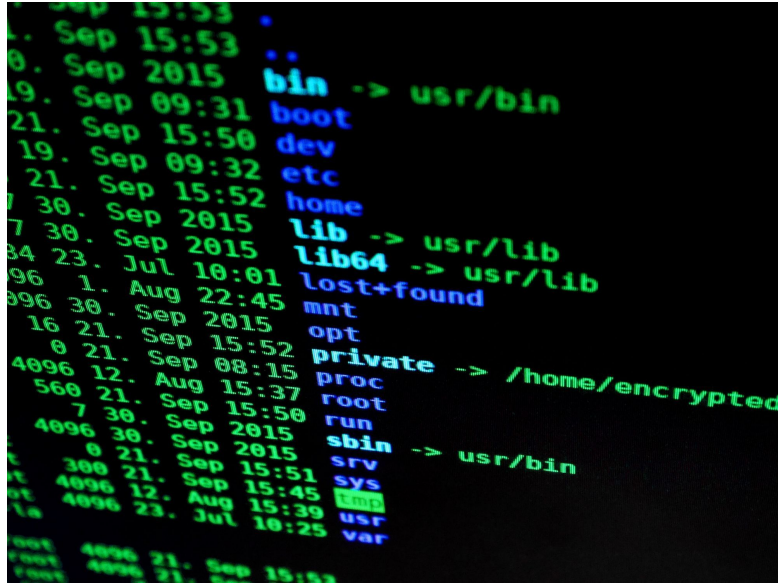
Poll: who is...

- Completely unfamiliar with using the CLI
- Somewhat familiar with it
- Very comfortable with it

The GUI



The Command Line



```
~/GitHub — -zsh — 160x48
/Users/tsmith > ls
total 0
drwxr-xr-x  5 tsmith staff 160 Sep 10 14:28 Code/
drwxr-xr-x  3 tsmith staff  96 Jan  4 09:30 'Creative Cloud Files/'
drwxr-xr-x 10 tsmith staff 320 Jan  6 16:53 Desktop/
drwxr-xr-x 12 tsmith staff 384 Nov 17 13:25 Documents/
drwxr-xr-x 15 tsmith staff 480 Jan  5 17:17 Downloads/
drwxr-xr-x 24 tsmith staff 768 Dec 31 11:10 GitHub/
drwxr-xr-x 80 tsmith staff 2.5K Dec 22 09:20 Library/
drwxr-xr-x  4 tsmith staff 128 Sep  4 09:33 Movies/
drwxr-xr-x  4 tsmith staff 192 Sep  4 09:33 Music/
drwxr-xr-x  4 tsmith staff 128 Mar 24 2020 OneDrive/
drwxr-xr-x  4 tsmith staff 128 Sep  3 15:51 Pictures/
drwxr-xr-x  4 tsmith staff 128 Jun 23 2017 Public/
drwxr-xr-x 15 tsmith staff 480 Dec 31 15:29 bin/
drwxr-xr-x  9 tsmith staff 288 Jan  6 15:27 img/
drwxr-xr-x 15 tsmith staff 480 Jul 22 17:07 impt-project/
drwxr-xr-x 11 tsmith staff 352 Sep 29 11:35 impt-test-project/
drwxr-xr-x 10 tsmith staff 320 Sep 17 14:05 yaml/
/Users/tsmith > pwd
/Users/tsmith
/Users/tsmith > cd GitHub
/Users/tsmith/GitHub > ls
total 0
drwxr-xr-x  6 tsmith staff 192 Dec 14 11:31 B096_GPS/
drwxr-xr-x 24 tsmith staff 768 Feb 17 2020 BLE/
drwxr-xr-x  8 tsmith staff 256 Jan 20 2020 BTLELinkUp/
drwxr-xr-x  6 tsmith staff 192 Aug 13 15:01 BlinkUpSOK-Android/
drwxr-xr-x  7 tsmith staff 224 Dec 31 14:25 BlinkUpSOK-iOS/
drwxr-xr-x 17 tsmith staff 544 Aug 19 17:11 'Connected Product'/
drwxr-xr-x 22 tsmith staff 704 Apr  2 2020 Experimental/
drwxr-xr-x  9 tsmith staff 288 Dec 31 16:53 Hex/
drwxr-xr-x  8 tsmith staff 256 Jan 27 2020 Legacy/
drwxr-xr-x  6 tsmith staff 192 Jun 30 2020 Onewire/
drwxr-xr-x 32 tsmith staff 1.0K Jan  5 13:46 Testing/
drwxr-xr-x  9 tsmith staff 288 Nov 23 09:31 cellular-iot/
drwxr-xr-x 14 tsmith staff 448 Aug 13 12:12 cordova-plugin-blinkup/
drwxr-xr-x 20 tsmith staff 640 Nov 30 09:26 doc-api/
drwxr-xr-x 11 tsmith staff 352 Jun 25 2020 dotfiles/
drwxr-xr-x 35 tsmith staff 1.1K Jul  3 2020 examples/
drwxr-xr-x 28 tsmith staff 896 Sep 17 16:24 junk-drawer/
drwxr-xr-x 12 tsmith staff 384 Jul  1 2020 mobile-android/
drwxr-xr-x 12 tsmith staff 384 Dec 31 13:48 mobile-ios/
drwxr-xr-x 10 tsmith staff 320 Jul 10 08:05 oktaupdate/
drwxr-xr-x 40 tsmith staff 1.3K Jan  4 10:05 scripts/
/Users/tsmith/GitHub >
```

Basic commands

- Navigation: `cd`, `ls`, `pwd`
- Filesystem ops: `cp`, `mv`, `rm`, `mkdir`
- File reading: `head`, `tail`, `cat`, `less`, `vim`
- Text manipulation: `cut`, `sort`, `uniq`, `awk`, `sed`, `tr`
- Search: `grep`, `find`
- Upload/Download: `wget`, `curl`, `scp`, `rsync`

Anatomy of a Command

user@machine:~/mydir \$ program subcommand -a -b --output results.txt input1.txt input2.txt

└── prompt ─┘ └cmd┘ └subcmd┘ └options┘ └ positional args ┘

- Common ways to specify values: -o value, --option value, --option=value
- Other ways: -o value, -o=value
- Short flags may be combined: ls -ltrh
- Help: man pages, -h/--help, cmd <no options>
- Spaces are delimiters
 - Quote values with spaces
 - If it also needs quotes, mix single & double quotes, or escape with \

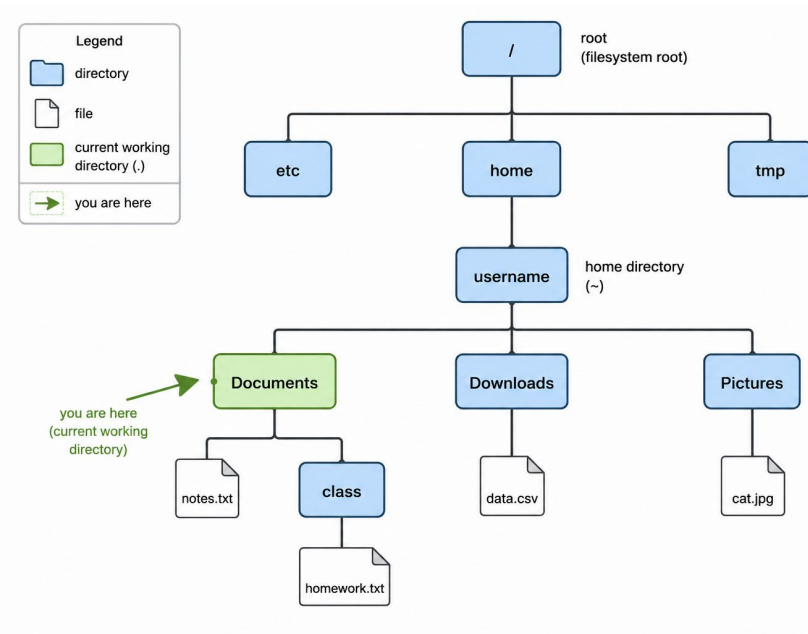
Filesystem & Paths

Absolute paths

- /home/username/Documents/notes.txt

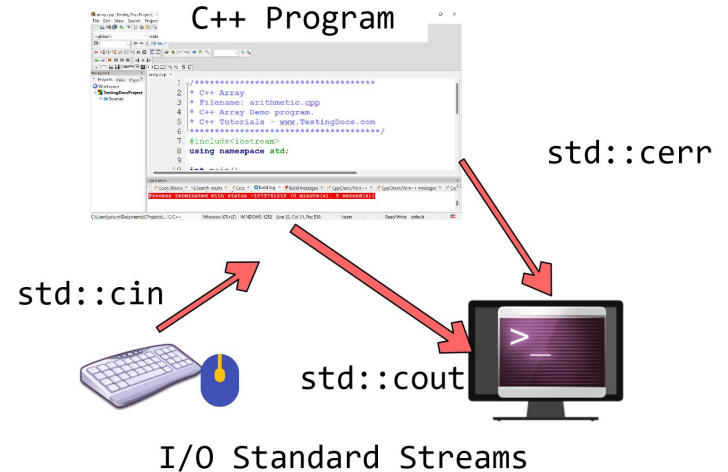
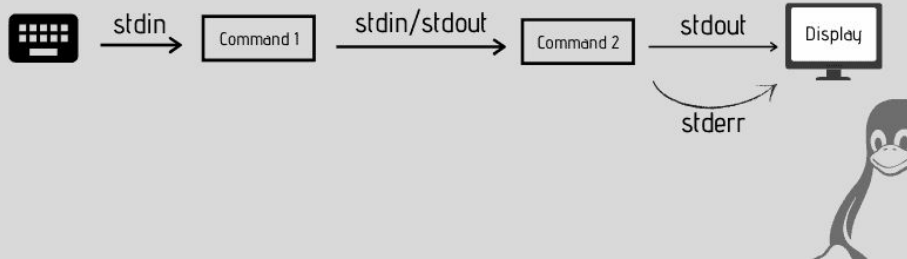
Relative paths

- notes.txt
- ./notes.txt
- class/homework.txt
- ../Downloads/data.tsv



File Streams

Redirection in Linux



Shell features: IO Redirection

Shells – Input/Output Redirection (3)

Operator	Description
<	Open the following file as stdin
>	Open the following file as stdout
>>	Append to the following file
<<del	Take stdin from here, up to the delimiter del
>&	Merge stdout with stderr
>>&	Append stdout to stderr
	Pipe stdout into stdin
n>&-	Close file descriptor

Most common:

- >
- >>
- |

Shell Features: Variables, Scripts, & More

- Variables: \$VAR
 - Special variables: \$HOME, \$PATH
- Globbing & Wildcards: *
- Loops
- Scripts
 - \ to continue a line
 - && to chain commands that succeed
- Tab autocomplete
- Escaping: ^C
- Remember: there are multiple ways to do things!

Basic commands

- Navigation: `cd`, `ls`, `pwd`
- Filesystem ops: `cp`, `mv`, `rm`, `mkdir`
- File reading: `head`, `tail`, `cat`, `less`, `vim`
- Text manipulation: `cut`, `sort`, `uniq`, `awk`, `sed`, `tr`
- Search: `grep`, `find`
- Upload/Download: `wget`, `curl`, `scp`, `rsync`